**Shaik Mahammad Khaleel**
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## SUMMARY

- Hands on experience in synthesis, STA, Floor Planning, Power Planning, Placement, CTS, Routing and IR Drop Analysis.
- Have solid experience in Logical and Physical Synthesis, Macro Placement, Routing at advanced Node, Static Timing Analysis, Timing Closure techniques, Noise analysis, Functional Eco's, Metal Eco's implementation.
- Experience in lower technology nodes such as TSMC 16nm, 28nm and 32nm able to handle complex designs with multimillion gates and tens of macros.
- Having better understanding of clock tree structure and implementation.
- Hands on experience in low power implementation techniques such as ON-OFF and Multi-Voltage designs.
- Have hands on experience in DRC, LVS, LEC, DRV fixes, Crosstalk analysis, Glitches, Antenna and EM/IR fixes.
- Strong interest and understanding of Advanced Node and Design methodologies.
- Proficient in scripting languages TCL, PERL and good at UNIX.

## WORK EXPERIENCE

- One year internship as a Physical Design Engineer at RiseTime semiconductors (OPC) Pvt Ltd, from 4th April, 2022 to 22nd April, 2023.

## ACADEMIC QUALIFICATIONS

| Qualification | Specialization | University/Board | Institute | Year | CGPA/Percentage |
|---|---|---|---|---|---|
| B.Tech | EEE | JNTUK | Tirumala Engineering college | 2020 | 8.34 CGPA |
| Diploma | EEE | AP SBTET | A.M. Reddy Memorial college | 2017 | 85.27% |
| Matriculation | SSC | Board of Secondary Education Andhra Pradesh | B R C high school | 2014 | 9.2 CGPA |

## TOOLS

| | |
|---|---|
| **Synthesis** | Design Compiler, Genus and Fusion Compiler |
| **Place and Route** | ICC2 and Fusion Compiler |
| **Timing Analysis** | Prime Time |
| **Logical Equivalent Check** | Formality |

## PROJECT DEATILS

**BLOCK I**

| | | | |
|---|---|---|---|
| Instance Count | : 2.2M | Number of Macros | : 16 |
| Technology node | : TSMC 28nm | Metal Layers | : 9 |
| Number of Clocks | : 2 | Operating Frequency | : 1.2 GHz, 520MHz |

Tools used : ICC2, PT, Formality.
Challenges faced : Congestion issues, local density, clock latencies & skew.
Role Played : Floorplanning, Place and Route, DRC clean, STA Signoff.

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**BLOCK II**

| | | | |
|---|---|---|---|
| Instance Count | : 2.8M | Number of Macros | : 186 |
| Technology node | : TSMC 16nmX9M | Operating Frequency | : 602MHz |
| Number of Clocks | : 4 | | |

Challenges faced : IR fixes
Tools used : ICC2, Prime Time, Formality.
Role Played : Performed Signoff Timing analysis, Calibre DRC & LVS and Formality checks.

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**BLOCK III**

| | | | |
|---|---|---|---|
| Instance Count | : 6.2k | Number of Macros | : 4 |
| Technology node | : TSMC 16nm | Metal Layers | : 9 |
| Number of Clocks | : 1 | Operating Frequency | : 150MHz |

Challenges faced : Congestion, fixing max cap, max trans, IR drop analysis DRC and LVS.
Tools used : ICC2, Prime Time, Formality.
Role Played : Sanity checks, Floorplanning, Placement, CTS, Routing, Timing analysis and Signoff.

## DECLARATION

I do hereby declare that the above particulars of facts and information stated are true, correct, and complete to the best of my belief and knowledge.

**Place :** Hyderabad **Signature**

**Date :** 23 April, 2023 Shaik Mahammad Khaleel